CPTG: Cosmological Scale Relation

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Abstract

In Curvature Polarization Transport Gravity, the Hubble scale H_0 is not treated only as an observational expansion parameter. It is a structured value tied directly to the internal acceleration scale $a_\star$. This relation connects galaxy-scale curvature response with the large-scale geometric expansion of the field.

1 Master Relation

CPTG defines the connection between the acceleration scale and the cosmological scale as

$$H_0 = \frac{16\pi}{3} \frac{a_\star}{c}. \tag{1}$$

Equivalently,

$$a_\star = \frac{3}{16\pi} c H_0. \tag{2}$$

This means that $a_\star$ and H_0 are not independent quantities. The acceleration scale measured in galaxy dynamics is structurally linked to the expansion scale of the theory.

2 Scale Coefficient

The coefficient relating $a_\star$ to cH_0 is

$$\frac{3}{16\pi} \approx 0.0597. \tag{3}$$

This lies very close to

$$\frac{3}{50} = 0.06. \tag{4}$$

In CPTG, this places the acceleration-expansion relation near a simple fractional lattice point.

Parameter	Value	Role
$a_\star$	$\approx 1.2 \times 10^{-10}$ m/s^2	galaxy acceleration scale
c	299,792,458 m/s	speed of light
$\frac{3}{16\pi}$	≈ 0.0597	CPTG scale coefficient
$\frac{3}{50}$	0.06	nearby lattice value

3 Derived Value of H_0

Using

$$a_\star \approx 1.2 \times 10^{-10}\ \mathrm{m/s}^2, \tag{5}$$

the CPTG relation gives

$$H_0 = \frac{16\pi}{3}\frac{a_\star}{c}. \tag{6}$$

This yields

$$H_0 \approx 207\ \mathrm{km/s/Mpc}. \tag{7}$$

This value is not interpreted as the same quantity measured by standard late-time distance-ladder methods or by CMB-based cosmological fits. In CPTG, it is the structured geometric expansion scale associated with curvature polarization and transport.

4 Difference from the Hubble Tension

The usual Hubble tension compares different observational estimates of cosmic expansion, such as lower CMB-inferred values and higher local distance-ladder values.

CPTG treats the issue differently. The CPTG value of H_0 is not merely a measurement of matter dilution through expanding space. It is a derived geometric field scale.

The distinction is:

$$\text{standard cosmology: } H_0 \to \text{observed matter-dilution expansion rate}, \tag{8}$$

while

$$\text{CPTG: } H_0 \to \text{structured geometric expansion scale}. \tag{9}$$

The larger CPTG value reflects the internal curvature scale of the theory. It represents the geometric expansion pressure associated with the same structure that produces the galaxy-scale acceleration $a_\star$.

5 Ratio to the Standard Expansion Scale

The CPTG geometric scale and the standard matter-based expansion scale are not unrelated. Comparing the two gives a simple fractional relation.

Using

$$H_{0,\text{standard}} \approx 67 \text{ km/s/Mpc} \tag{10}$$

and

$$H_{0,\text{CPTG}} \approx 207 \text{ km/s/Mpc}, \tag{11}$$

the ratio is

$$\frac{67}{207} \approx 0.323. \tag{12}$$

This is close to

$$\frac{16}{50} = 0.32. \tag{13}$$

In the CPTG lattice, 16/50 is the locked shell ratio used in early curvature-transport structure. This suggests that the standard expansion scale may represent a locked sub-mode of the full CPTG geometric expansion scale.

In this interpretation,

$$H_{0,\text{standard}} \approx \frac{16}{50} H_{0,\text{CPTG}}. \tag{14}$$

The observed expansion rate is therefore not rejected. It is interpreted as a projected or locked component of the larger geometric field scale.

6 Inverse Ratio

The inverse comparison gives

$$\frac{207}{67} \approx 3.09. \tag{15}$$

This lies near the lattice values

$$\frac{154}{50} = 3.08 \tag{16}$$

and

$$\frac{155}{50} = 3.10. \tag{17}$$

It is also close to π, which already appears in the CPTG master relation:

$$H_0 = \frac{16\pi}{3} \frac{a_\star}{c}. \tag{18}$$

This connects the expansion ratio back to the geometric structure of the theory. The factor of π governs the transition from a linear acceleration scale to a spherical curvature-polarization scale.

7 Ablation Insight

The ratio

$$\frac{67}{207} \approx 0.323 \tag{19}$$

is close to the locked shell value

$$\frac{16}{50} = 0.32. \tag{20}$$

If this ratio is shifted significantly, the connection to the 16/50 shell is lost. In CPTG, this is important because the standard expansion scale and the CPTG geometric expansion scale are produced by different measurements, yet their ratio falls near a strong lattice point.

This suggests that the Hubble tension is not simply a measurement conflict. In CPTG, it may reflect a geometric ratio between the observed matter-dilution expansion and the full curvature-field expansion scale.

8 Physical Meaning

The relation

$$a_\star = \frac{3}{16\pi} cH_0 \tag{21}$$

removes the apparent coincidence between the galactic acceleration scale and the cosmological expansion scale.

In CPTG, the connection is structural:

$$a_\star \sim cH_0. \tag{22}$$

The acceleration scale appearing in galaxies and the expansion scale appearing cosmologically are two expressions of the same curvature framework.

The standard value near 67 km/s/Mpc is then interpreted as a locked observational sub-mode of the larger CPTG geometric scale near 207 km/s/Mpc.

9 Summary

CPTG links local galactic dynamics and cosmological expansion through a fixed structural relation:

$$H_0 = \frac{16\pi}{3} \frac{a_\star}{c}. \tag{23}$$

Using

$$a_\star \approx 1.2 \times 10^{-10}\ \mathrm{m/s^2}, \tag{24}$$

this gives

$$H_0 \approx 207\ \mathrm{km/s/Mpc}. \tag{25}$$

This is interpreted as a CPTG geometric expansion scale rather than a direct replacement for standard observational Hubble measurements.

The ratio between the standard expansion value and the CPTG geometric value is

$$\frac{67}{207} \approx 0.323 \approx \frac{16}{50}. \tag{26}$$

Thus, the Hubble tension becomes a structural relation: the observed expansion scale is a locked fraction of the full CPTG geometric expansion scale.